

# IRON HAVEN

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## Safety Intelligence

Real-Time Underground Tunnel Safety Monitoring System

### PRESENTED BY

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### YEAR

2026

## MOTIVATION

# WHY THIS PROJECT EXISTS.

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On July 2, 2026, the Teesta Stage-VI hydroelectric tunnel in Sikkim collapsed, trapping over 30 workers underground. Rescue teams arrived with no sensor data, no communication channel, and no way to locate workers inside the tunnel.

India has 200+ active tunnel projects across highways, railways, and hydroelectric infrastructure — yet not a single one uses a digital safety monitoring system. Workers rely on hand signals and walkie-talkies.

The workforce speaks 20+ languages. Safety manuals in English are useless to a Santali miner from Jharkhand or a Meitei worker from Manipur. Any real solution must speak the worker's own language.

**200+**

Active tunnel  
projects

**50-80**

Worker deaths  
per year

**ZERO**

Digital systems  
deployed

**20+**

Languages  
spoken

## THE SOLUTION

# IRONHAVEN.

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A single HTML file that turns any browser into a full-scale underground tunnel safety command center. Real-time sensor monitoring, two-way worker-admin communication, 23 languages, audio sirens — zero installation, zero infrastructure, zero dependencies.

### INSTANT

Opens in any browser  
No install needed

### 23 LANGUAGES

All scheduled languages  
of India + Tibetan

### 9 SENSORS

Updating every  
3 seconds

### AUDIO ALERTS

Automatic danger  
sirens via Web Audio

### TWO-WAY COMMS

Worker reports to  
Admin instructions

METHODOLOGY

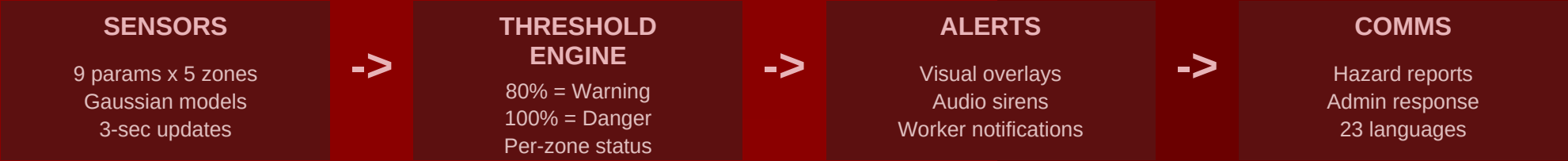
# HOW IT WAS BUILT.

|              |                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESEARCH     | Studied real tunnel disaster reports — Teesta Stage-VI (2026), Silkyara (2023) — to map exact failure points and identify 9 critical environmental parameters from DGMS standards. |
| SIMULATION   | Designed Gaussian distribution models (Box-Muller transform) for each sensor with calibrated mean, std dev, and danger thresholds from real tunnel data.                           |
| ARCHITECTURE | Single-file HTML + CSS + JS for maximum portability. Custom SVG sparklines, Web Audio API sirens, MediaRecorder voice/photo capture — all native browser APIs.                     |
| i18n         | Compiled 23-language translation dictionary. All 22 Scheduled Languages of India + Tibetan (Bhotia), rendered in native scripts.                                                   |
| ITERATION    | Built across 23 published versions. Each version addressed feedback — sensor tuning, communication flow, language accuracy, and edge cases.                                        |

# ARCHITECTURE OVERVIEW.

| Path                            | Description                                     | Details          |
|---------------------------------|-------------------------------------------------|------------------|
| public/index.html               | Complete dashboard — HTML+CSS+JS self-contained | 3,311 lines      |
| data/tunnel_monitoring_5000.csv | Synthetic dataset for ML model training         | 5,000 rows       |
| notebooks/Ironhaven_ML.ipynb    | ML classification and anomaly detection         | Python / sklearn |
| vercel.json                     | Deployment config for Vercel static hosting     | 4 lines          |
| README.md                       | Full project documentation                      | Markdown         |

## DATA FLOW



SENSOR ENGINE

# 9 SENSORS. 5 ZONES.

Each sensor runs a Gaussian simulation (Box-Muller transform) calibrated from real tunnel safety data. Readings update every 3 seconds.

| Sensor          | Unit  | Mean +/- Std Dev  | Danger    |
|-----------------|-------|-------------------|-----------|
| Temperature     | C     | 30.04 +/- 5.02    | 38 C      |
| Carbon Monoxide | ppm   | 19.96 +/- 14.66   | 35 ppm    |
| Methane         | %LEL  | 2.91 +/- 2.13     | 5% LEL    |
| Smoke Level     | ug/m3 | 36.05 +/- 27.19   | 70 ug/m3  |
| Humidity        | %RH   | 64.89 +/- 9.89    | 85% RH    |
| AQI             | --    | 102.84 +/- 74.45  | 200       |
| Vibration       | g     | 0.05 +/- 0.03     | 0.1 g     |
| CO2 Level       | ppm   | 509.59 +/- 279.89 | 1000 ppm  |
| Light Intensity | lux   | 251.44 +/- 79.2   | < 100 lux |

| Zone | Name            | Capacity     | Primary Hazard      |
|------|-----------------|--------------|---------------------|
| A    | Intake Shaft    | 4-8 workers  | Poor ventilation    |
| B    | Main Bore       | 8-15 workers | Gas accumulation    |
| C    | Deep Excavation | 6-12 workers | Structural collapse |
| D    | Tunnel Face     | 5-10 workers | Blast overpressure  |
| E    | Ventilation Bay | 3-6 workers  | Heat exhaustion     |

## KEY FEATURES

# WHAT THE SYSTEM DOES.

### ADMIN DASHBOARD

Live 5x9 sensor grid, configurable thresholds, worker zone map, hazard report inbox, alert feed, emergency contacts.

### ALERT SYSTEM

Three tiers — SAFE (green), WARNING at 80% (amber), DANGER at 100% (red overlay + dual-tone audio siren).

### COMMUNICATION

Worker -> Hazard Report (9 types, voice + photo) -> Admin Inbox -> Reply in worker's language -> Acknowledged.

### WORKER PORTAL

Language selector (23 languages), zone status, safety gear checklist, hazard report form with voice + photo, admin messages.

### MULTILINGUAL

All 22 Scheduled Languages of India + Tibetan (Bhotia). Every label, button, alert renders in native script.

### PERSISTENCE

Thresholds, contacts, language, theme persist in localStorage. 5 default emergency contacts + custom. Dark/light toggle.

### MANDATORY SAFETY GEAR (ALL ZONES):

Hard Hat (Class E) | Safety Goggles | Respirator (N95/P100) | Hi-Vis Vest (Class 3)  
Cut-Resistant Gloves | Steel-Toe Boots | Ear Protection (NRR 25+) | Two-Way Radio

TECHNOLOGY STACK

# ZERO LIBRARIES. PURE WEB.

The entire application — charts, audio, internationalization, real-time updates — uses native browser APIs only. No npm, no CDN, no dependencies.

| Layer      | Technology                                      | Why It Matters         |
|------------|-------------------------------------------------|------------------------|
| Frontend   | HTML5 + CSS3 (Custom Properties, Grid, Flexbox) | Zero build step        |
| Charting   | Custom SVG sparkline renderer                   | No Chart.js dependency |
| Audio      | Web Audio API (OscillatorNode + GainNode)       | No audio files needed  |
| Media      | MediaRecorder API + FileReader API              | Voice + photo capture  |
| Storage    | localStorage API                                | Settings persist       |
| Simulation | Gaussian random (Box-Muller transform)          | Realistic sensor noise |
| Theming    | CSS Custom Properties + data-theme              | Dark/light toggle      |
| Deploy     | Vercel static hosting                           | Global CDN, auto-SSL   |

3,311

Lines of code

23

Versions published

9

Live sensors

23

Languages

12

Workers tracked

0

External libraries



# LIMITATIONS & ROADMAP.

|                       |                                             |                                     |
|-----------------------|---------------------------------------------|-------------------------------------|
| <b>Simulated Data</b> | Sensors run Gaussian models, not real IoT   | -> MQTT broker + real sensors       |
| <b>Single Browser</b> | No backend — data in one tab, one device    | -> Node.js / FastAPI + WebSocket    |
| <b>No Auth</b>        | No login — anyone can access admin view     | -> RBAC (Admin, Supervisor, Worker) |
| <b>No GPS</b>         | Zone-based simulation, not real positioning | -> UWB sensors or BLE beacons       |

## FUTURE ROADMAP

|                           |                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------|
| <b>PHASE 1 (3-6 mo)</b>   | Real IoT via MQTT   Multi-device WebSocket sync   Auth with RBAC   PWA (offline -first) |
| <b>PHASE 2 (6-12 mo)</b>  | ML anomaly detection (LSTM)   Predictive alerts   CV-based PPE compliance               |
| <b>PHASE 3 (12-24 mo)</b> | Multi-tunnel command center   BIM 3D visualization   Regulatory reporting               |

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*"Every tunnel deserves real-time awareness.  
Every worker deserves safety in their own language.  
Every administrator deserves data that saves lives."*

## CREATOR

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